

Jacobsthal Slide and Shiftless Orbit in Collatz Trajectory

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Abstract

This report introduces the Shiftless Model, which reinterprets the “ $3n + 1$ ” operation and the unpredictable number of divisions by 2 in the Collatz mapping as the adjacent sum (interference) and the superposition of the basic wave (slide) in Jacobsthal (J) space. By taking the adjacent sum within the J-wave to which an odd kernel n belongs, a pure shift stream 2^k is extracted. Superimposing the basic J-wave $J(0,1)$ according to the hierarchy then generates the J-wave of the next odd kernel n' . This operation, called the **J Slide**, allows the Collatz trajectory to be interpreted geometrically as Wave Switching, yielding a **Shiftless Orbit** that eliminates the uncertainty of division by 2.

1. Introduction

The difficulty of the Collatz conjecture lies in the inability to predict how many times an odd number n will be divided by 2 after the $3n + 1$ operation. This study presents a model that separates and absorbs this “shift uncertainty” using the structure of the Jacobsthal (J) sequence, revealing a geometric and wave-based interpretation of the Collatz trajectory.

2. The Shiftless Mechanism

2.1 The Basic J-wave $J(0, 1)$

The J-sequence is defined by the following recurrence relation:

$$J_{k+2} = J_{k+1} + 2J_k$$

The basic J-wave $J(0,1)$ begins as:

$$J_k = [0, 1, 1, 3, 5, 11, \dots]$$

A key property of this sequence is that the adjacent sum always yields a pure power of 2:

$$J_k + J_{k+1} = 2^k \quad (k \geq 1)$$

$$1 + 1 = 2^1, 1 + 3 = 2^2, 3 + 5 = 2^3, 5 + 11 = 2^4$$

This property corresponds exactly to the “shift stream” (division by 2) in the Collatz mapping.

2.2 Interference and Superposition in J-wave

Consider a J-wave $J(a, b)$ to which an arbitrary odd kernel n belongs:

$$J(a, b) = [a, b, 2a + b, 2a + 3b, 6a + 5b, \dots]$$

Taking the adjacent sum cancels the J-components and extracts a pure shift stream:

$$Adjacent\ Sum = [a + b, 2(a + b), 4(a + b), 8(a + b), \dots]$$

Next, superimposing the basic J-wave $J(0,1)$ according to the hierarchy k generates the J-wave of the next odd kernel:

$$J_{next} = [a + b + 0, 2(a + b) + 1, 4(a + b) + 1, 8(a + b) + 3, \dots]$$

This operation is the **J Slide**. At each hierarchy, the **Least Significant Jacobsthal (LSJ)** is added, causing the odd kernel to advance while switching J-waves.

3. Verification: Orbit 7 → 1

In this report, the operation of dividing by 2 as many times as possible after $3n + 1$ is called a Minimized Shift. This produces the odd kernel of the conventional Collatz odd step. The odd Collatz trajectory starting from 7 is:

$$7 \rightarrow 11 \rightarrow 17 \rightarrow 13 \rightarrow 5 \rightarrow 1$$

For each node, **Table 1** shows the expansion of the J-spinor wave and the shiftless generation of the next odd kernel.

Table 1. Shiftless Collatz Orbit of Initial Value 7

Level	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3
J2(0,1)	-5461/16384	2731/8192	-1365/4096	683/2048	-341/1024	171/512	-85/256	43/128	-21/64	11/32	-5/16	3/8	-1/4	1/2	0	1	1	3

Level	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3
J2 Expansion	-2727/8192	1369/4096	-679/2048	345/1024	-167/512	89/256	-39/128	25/64	-7/32	9/16	1/8	5/4	3/2	4	7	15	29	59
Marge	11/8192	11/4096	11/2048	11/1024	11/512	11/256	11/128	11/64	11/32	11/16	11/8	11/4	11/2	11	22	44	88	176
J2 Slide	1371/4096	-677/2048	347/1024	-165/512	91/256	-37/128	27/64	-5/32	11/16	3/8	7/4	5/2	6	11	23	45	91	181
Marge	17/4096	17/2048	17/1024	17/512	17/256	17/128	17/64	17/32	17/16	17/8	17/4	17/2	17	34	68	136	272	544
J2 Slide	-337/1024	175/512	-81/256	47/128	-17/64	15/32	-1/16	7/8	3/4	5/2	4	9	17	35	69	139	277	555
Marge	13/1024	13/512	13/256	13/128	13/64	13/32	13/16	13/8	13/4	13/2	13	26	52	104	208	416	832	1664
J2 Slide	-41/128	23/64	-9/32	7/16	-1/8	3/4	1/2	2	3	7	13	27	53	107	213	427	853	1707
Marge	5/128	5/64	5/32	5/16	5/8	5/4	5/2	5	10	20	40	80	160	320	640	1280	2560	5120
J2 Slide	3/8	-1/4	1/2	0	1	1	3	5	11	21	43	85	171	341	683	1365	2731	5461
Marge	1/8	1/4	1/2	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384
J2 Slide	1/2	0	1	1	3	5	11	21	43	85	171	341	683	1365	2731	5461	10923	21845

3.1 Step 1: Node 7

J-spinor: $J(4,7)$

J-wave: 4, 7, 15, 29, 59, ...

Interference: 11, 22, 44, 88, ...

Slide: 11, 23, 45, 91, ...

Minimized Shift: 6, 11, 23, 45, 91, ...

3.2 Step 2: Node 11

J-spinor: $J(6,11)$

J-wave: 6, 11, 23, 45, 91, ...

Interference: 17, 34, 68, 136, ...

Slide: 17, 35, 69, 139, ...

Minimized Shift: 4, 9, 17, 35, 69, 139, ...

3.3 Step 3: Node 17

J-spinor: $J(4,9)$

J-wave: 4, 9, 17, 35, 69, ...

Interference: 13, 26, 52, 104, ...

Slide: 13, 27, 53, 107, ...

Minimized Shift: 2, 3, 7, 13, 27, 53, 107, ...

3.4 Step 4: Node 13

J-spinor: $J(2,3)$

J-wave: 2, 3, 7, 13, 27, ...

Interference: 5, 10, 20, 40, ...

Slide: 5, 11, 21, 43, ...

Minimized Shift: 0, 1, 1, 3, 5, 11, 21, 43, ...

3.5 Step 5: Node 5

J-spinor: $J(0,1)$

J-wave: 0, 1, 1, 3, 5, ...

Interference: 1, 2, 4, 8, ...

Slide: 1, 3, 5, 11, ...

Minimized Shift: 0, 1, 1, 3, 5, 11, ...

The trajectory ultimately converges to the basic J-wave $J(0,1)$.

3.6 Shiftless Orbit

The Level 0 vertical cross-section of **Table 1** yields:

$$7 \rightarrow 23 \rightarrow 69 \rightarrow 213 \rightarrow 683 \rightarrow 2731$$

No division by 2 occurs here. Each node switches J-waves while maintaining the hierarchy of the initial value.

$$\begin{aligned} 7 &\xrightarrow{\text{Adjacent Sum } 7+15=22} 22 \xrightarrow{+LSJ(1)} 23 \\ 23 &\xrightarrow{\text{Adjacent Sum } 23+45=68} 68 \xrightarrow{+LSJ(1)} 69 \\ 69 &\xrightarrow{\text{Adjacent Sum } 69+139=208} 208 \xrightarrow{+LSJ(5)} 213 \\ 213 &\xrightarrow{\text{Adjacent Sum } 213+427=640} 640 \xrightarrow{+LSJ(43)} 683 \\ 683 &\xrightarrow{\text{Adjacent Sum } 683+1365=2048} 2048 \xrightarrow{+LSJ(683)} \mathbf{2731} \end{aligned}$$

The final J-component 2731 matches J_{13} of the basic J-wave $J(0,1)$, suggesting that any odd Collatz trajectory is ultimately absorbed into the basic J-wave in J-space.

4. Conclusion

This study demonstrates that the Collatz mapping can be explained by the simple structure of adjacent sum (interference) and superposition of the basic J-wave (slide) inherent in J-waves. By eliminating the arithmetic uncertainty of division by 2 and interpreting the trajectory as a Shiftless Orbit that preserves hierarchy, the convergence to 1 becomes a natural return to the basic J-wave through Wave Switching.

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